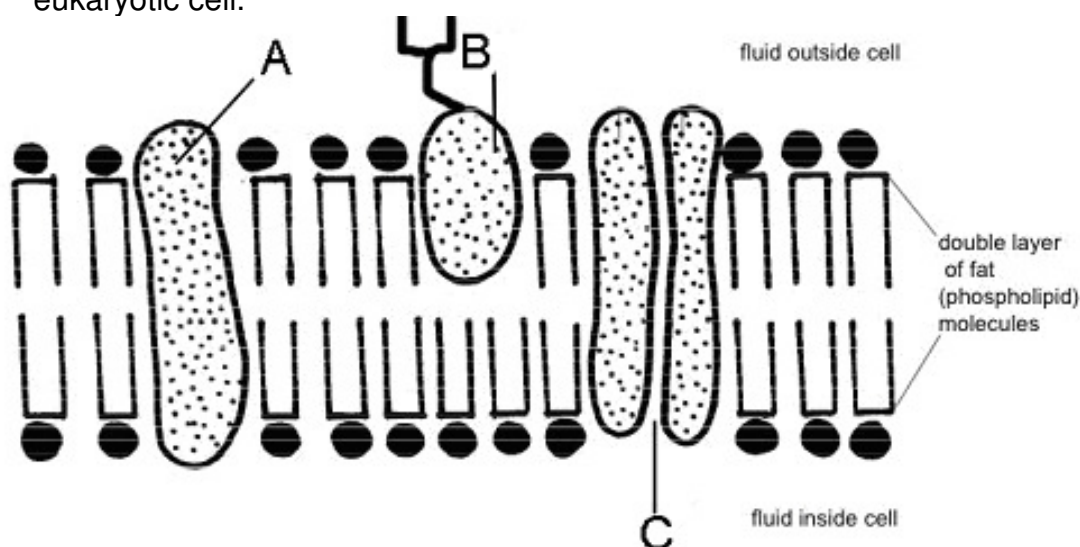


1. Some cellular organelles are bound by a single membrane, while other organelles have two membrane envelopes around them. Which of the following is correct?

	single membrane		two membranes	
A	tonoplast	lysosome	nucleus	chloroplast
B	chloroplast	lysosome	nucleus	tonoplast
C	nucleus	lysosome	chloroplast	tonoplast
D	tonoplast	chloroplast	nucleus	lysosome

2. The diagram shows a simplified model for the plasma membrane of an eukaryotic cell.



(adapted from Wikipedia)

Which option shows the most likely scenario for the three labeled proteins?

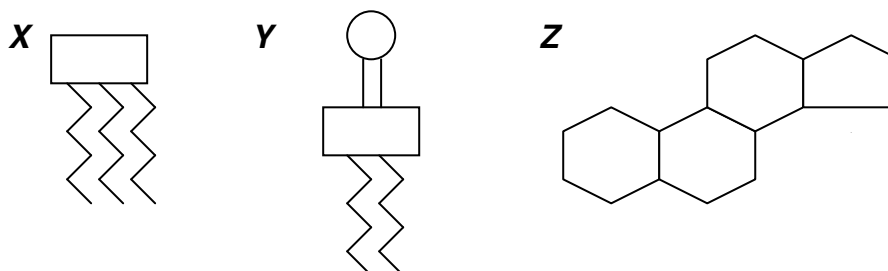
	protein A	protein B	protein C
A	glycoprotein	calcium channel	transmembrane protein
B	transmembrane protein	glycoprotein	chloride channel
C	glycoprotein	glycolipid	sodium channel
D	cytoskeleton	glycoprotein	potassium channel

3. Raffinose is a trisaccharide which can be degraded by using enzymes. The results of two different enzymic incubations are shown in the table.

Enzyme used	Products
invertase	melibiose + fructose
galactosidase	galactose + sucrose

Which one of the following statements is not consistent with this information?

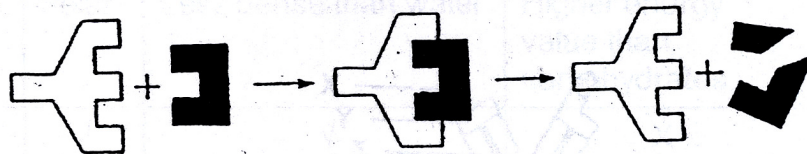
- A Melibiose is a disaccharide
- B Raffinose is composed of three different monosaccharides.
- C Acid hydrolysis of raffinose would yield some glucose
- D Raffinose would move further from the origin than melibiose during paper chromatography.
4. Which of the following illustrates hydrolysis?
- A The reaction of two monosaccharides to form a disaccharide with the release of water.
- B The synthesis of two amino acids to form a dipeptide with utilization of water.
- C The synthesis of nucleotide from a phosphate, a ribose sugar, and a nitrogenous base with the production of a molecule of water
- D The reaction of a fat to form glycerol and fatty acids with the utilization of water.
5. The figures below represent lipid and lipid related molecules.



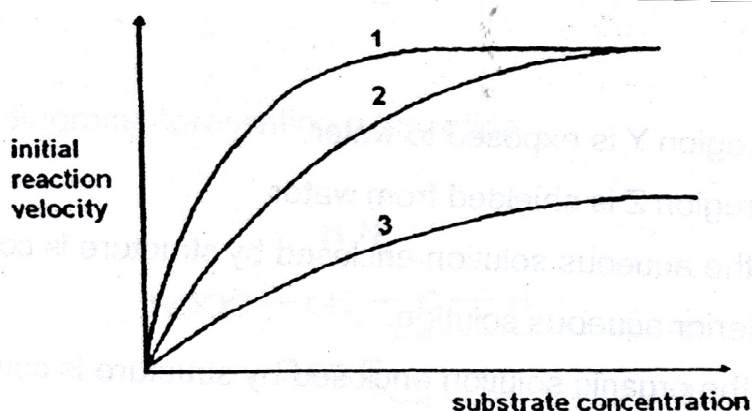
Name the above molecules.

	X	Y	Z
A	triglyceride	phospholipid	steroid
B	phospholipid	fatty acid	glycerol
C	steroid	triglyceride	fatty acid
D	fatty acid	phospholipid	steroid

6. The diagram below shows how an enzyme will act on its substrate.



The rates of reaction of this enzyme were determined in the presence and absence of inhibitors. The curves obtained are shown below.



What do the curves represent?

	Curve 1	Curve 2	Curve 3
A			
B			
C			
D			

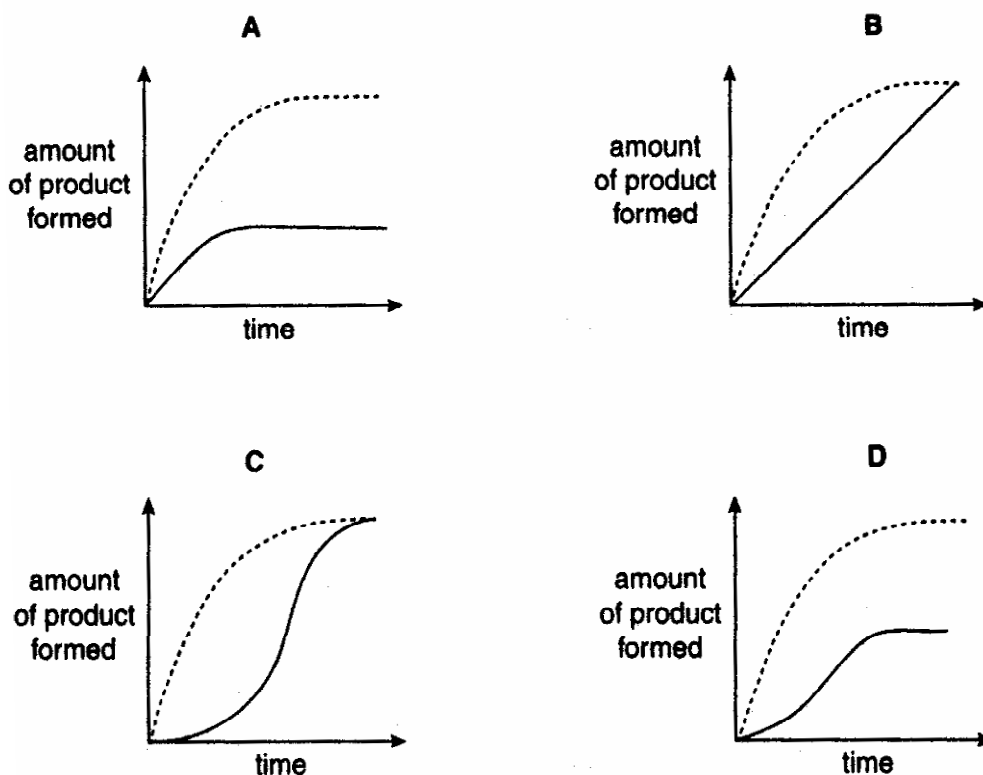
7. A constant amount of enzyme is added to a constant amount of substrate. The amount of product formed is measured over a period of time.

Which graph shows the effect of adding a small amount of a non-competitive inhibitor to the reaction mixture at the start?

key

----- uninhibited reaction

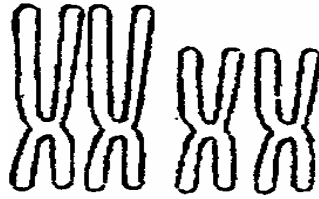
—— inhibited reaction



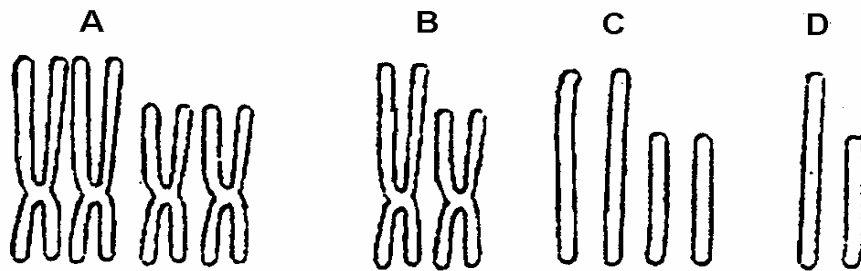
8. A bacterial amylase was incubated with starch at 20°C and pH 7. When a sample was removed after 5 minutes and mixed with iodine reagent the mixture was light brown colour. The most likely explanation of this observation is that

- A an inhibitor of the amylase was present
- B the pH was too high to allow the reaction to proceed quickly
- C the starch in the reaction mixture had been converted to maltose
- D bacterial amylase does not catalyse the digestion of starch

9. The diagram shows the chromosome of a cell just before the metaphase stage of mitosis.



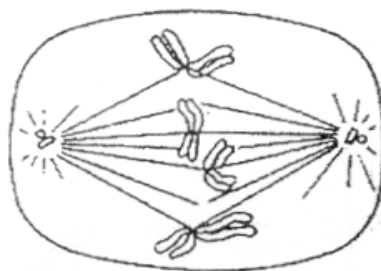
What will be the appearance of the chromosomes at telophase?



10. If X units of DNA are present in the nucleus of a cell, which has just divided, what is the relative amount of DNA present in this cell during anaphase of the next mitosis?

- A $X/4$
- B $X/2$
- C X
- D $2X$

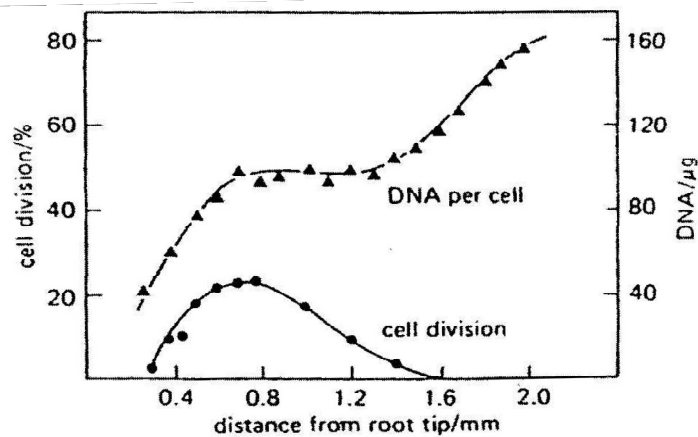
11. The cell below shows 4 chromosomes during nuclear division.



Which of the following correctly describes the cell?

	Type of nuclear division	Stage of nuclear division	Number of copies of chromosomes
A	mitosis	metaphase	2
B	meiosis	metaphase	2
C	mitosis	anaphase	4
D	meiosis	anaphase	4

12. The graph below shows the changes in DNA content and rate of division in cells at varying distances from the tip of an onion root.



Which one of the following can be determined from the graph?

- A Cells nearest the tip are dividing by meiosis.
 - B DNA quantity varies inversely with the cell division rate.
 - C DNA synthesis precedes cell division
 - D The cells are becoming polyploidy.
13. A particular sequence in one of the DNA strands (strand that is complementary to the strand used as a template for transcription) reads CAT-CGG.

Which one of the following shows the corresponding sequence on the anti-codon?

- A GTA-GCC
 - B GGC-AUG
 - C CAU-CGG
 - D GUA-GCC
14. Identify the correct statements regarding the replication of DNA
- 1 Only 1 strand of DNA is replicated
 - 2 It occurs in S phase of interphase in both mitosis and meiosis
 - 3 It occurs after the completion of telophase in meiosis
 - 4 DNA polymerase III is required (for bacteria).

- A 1 and 3 only
- B 2 and 4 only
- C 2 and 3 only
- D 2 only

15. The diagram shows a gene in a human cell



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

- A activation of the gene by a steroid hormone
 - B attachment of mRNA to a ribosome
 - C binding of RNA polymerase to DNA
 - D splicing of pre-mRNA
16. Four different genes are expressed in different ways.
- Gene 1** undergoes tissue-specific patterns of alternative splicing
- Gene 2** is part of a group of structural genes controlled by the same regulatory sequences
- Gene 3** is in some circumstances subject to methylation
- Gene 4** codes for a repressor protein which acts at an operator site close by
- Which of the above genes are of eukaryotic origin?
- A 1 only
 - B 1 and 3 only
 - C 2 and 3 only
 - D 2, 3 and 4 only
17. Which of the listed features are characteristics of an eukaryotic genome?

- 1 circular DNA
- 2 DNA associated with histone
- 3 DNA complexed to form nucleosome
- 4 DNA may be transferred to new cells by mitosis

- A 1 only
- B 1 and 3 only
- C 2 and 3 only
- D 2, 3 and 4 only

18. A color blind man (who is otherwise normal) marries a woman who has normal vision but is hemophilic. Given that both these traits are X-linked and recessive, which of the following can be expected from their offspring

	sons		daughters	
A	50% hemophilic, normal vision	50% non-hemophilic, normal vision	100% hemophilic and color blind	
B	100% normal vision and hemophilic		100% carriers for color blindness and hemophilic	
C	100% carriers for hemophilia and color blindness		100% carriers for color blindness	
D	50% carriers for hemophilia	50% normal	50% carriers for color blindness	50% carriers for hemophilia

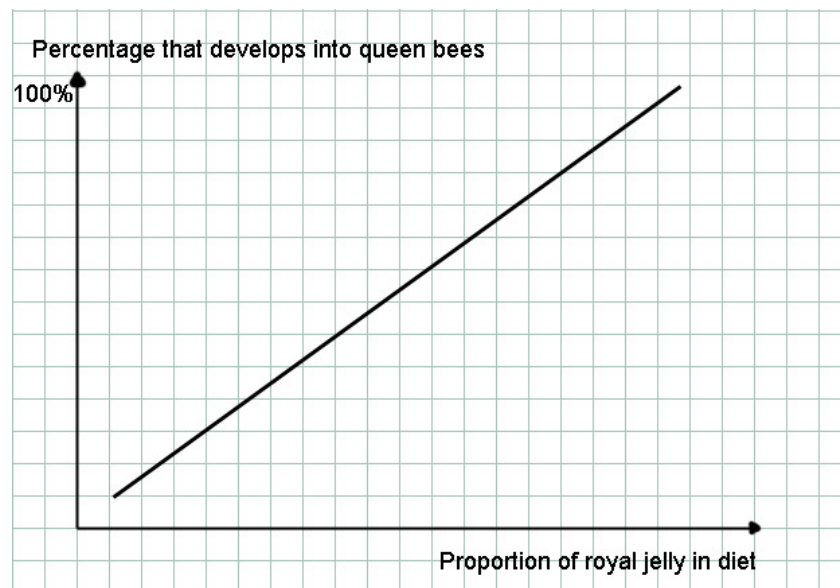
19. A man with **O** blood type married a woman with **AB** blood type and they have four children in the family. The first has **A** blood type, the second has **B** blood type, the third has **AB** blood type and the fourth has **O** blood type.

Which of the following conclusions is possible from the information given?

- A** The first two children are adopted as they do not share the blood type of their parents
- B** The third and fourth children are probably adopted as the parents cannot produce children with these phenotypes
- C** Only the third child belonged to the parents and the others are probably adopted as the **A** and **B** allele are codominant
- D** It is possible that all four children are the offspring of the pair of parents because of independent assortment of chromosomes
20. Which of the following are true of continuous and discontinuous variation?

	continuous variation	discontinuous variation
A	many intermediates, controlled by several genes	few intermediates, controlled by one or two genes
B	few intermediates, controlled by one or two genes	many intermediates, controlled by several genes
C	few intermediates, controlled by several genes	many intermediates, controlled by one or two genes
D	many intermediates, controlled by one or two genes	few intermediates, controlled by several genes

21. Analyse the following graph about worker honeybees fed on royal jelly and select the best conclusion.



- A Worker bees develop into queen bees only when fed exclusively on royal jelly.
 - B Worker bees develop into queen bees over time.
 - C Having a higher proportion of royal jelly in the diet causes genetic changes in honeybees.
 - D The phenotype of the honeybee is dependent on diet.
22. During aerobic respiration, ATP can be formed by oxidative phosphorylation in the electron transport chain and by glycolysis.

In complete oxidation of one molecule of glucose, approximately what percentage of ATP is formed by oxidative phosphorylation?

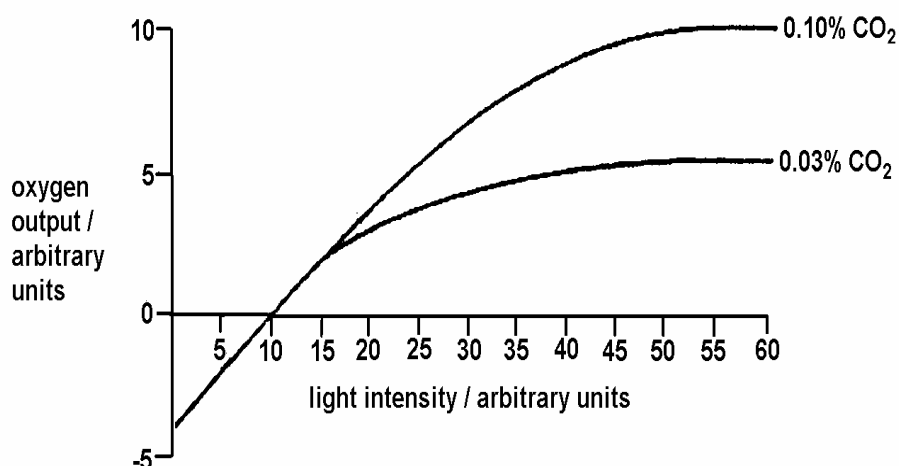
- A 10%
- B 25%
- C 75%
- D 90%

23. Removal of the source of carbon dioxide from photosynthesizing chloroplasts results in rapid changes in the concentration of certain chemicals.

Which one of the following represents the correct combination of concentration changes?

	ATP	Ribulose diphosphate (bisphosphate)	Phosphoglyceric acid (PGA)
A	decreases	decreases	increases
B	decreases	increases	no change
C	increases	increases	decreases
D	increases	no change	increases

24. The graph shows oxygen exchange in a green plant at different light intensities and at two concentrations of carbon dioxide in the surrounding air.



At which light intensity are photosynthesis and respiration occurring at the same rate?

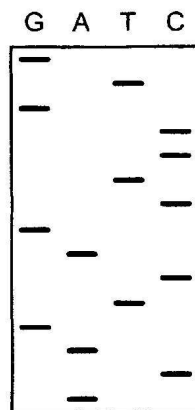
- A** 0
- B** 10
- C** 15
- D** 40

25. A large population of equal numbers of dark and light mice was released into an area where owls are predators of mice. Because of predation, only 25% of these mice survived and selective killing by owls changed the proportions of dark to light mice from 1:1 to 4:1.

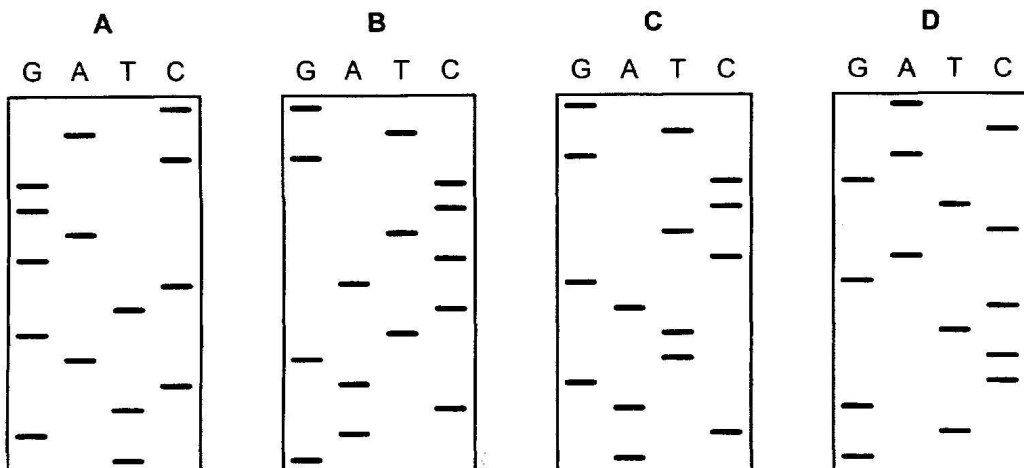
If mice produce an average of 8 offspring per litter and the pattern of predation remains the same, what would happen to the population of light mice?

- A They disappear after one further generation.
 - B They disappear after two further generations.
 - C They remain at the level of one fifth of the population.
 - D They would be reduced to a constant but very low frequency.
26. In people with a mutant allele, a protein contains just one different amino acid in the primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using electrophoresis.

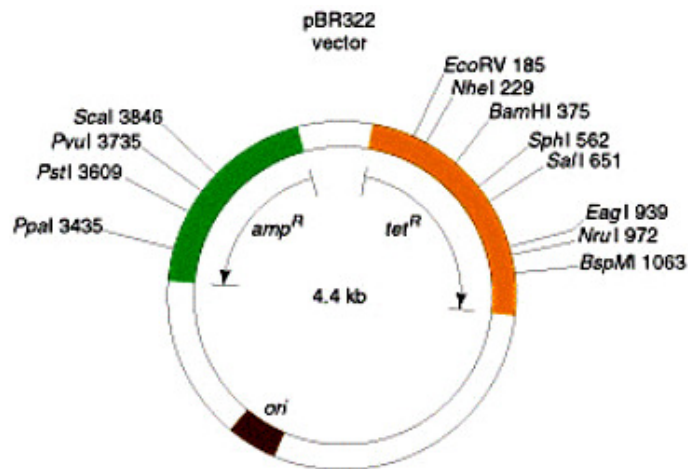
The electrophoresis result from the DNA of a normal allele of a gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?



27. pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction enzyme *Bam*HI.



Following transformation, bacterial cells were grown in four different media, as shown below:

I	Nutrient broth plus ampicillin
II	Nutrient broth plus tetracycline
III	Nutrient broth plus ampicillin and tetracycline
IV	Nutrient broth without antibiotics

Which of the following media would bacterial cells that contain the recombinant plasmids grown in?

- A** I and II
B I and III
C I and IV
D IV only
28. Which of the following characteristics is undesirable in cloning vectors used in genetic engineering?
- A** Control their own replication
B High copy number
C Small in size
D Vulnerable at several sites to a restriction enzyme

- 29.** Which statement correctly describes pluripotency?
- E** The ability to develop into a whole organism
 - F** The ability to develop into tissues from various cell types
 - G** The ability to develop into different cells of a particular tissue type
 - H** The ability to stay undifferentiated
- 30.** Which process could produce a large number of genetically identical plants?
- A** pollen culture
 - B** embryo culture (plantlet)
 - C** meristematic tissue culture (protoplast)
 - D** anther culture